

Ground Effects on the Vortex-induced Vibration of Bridge Decks

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Received March 11, 2018/Revised August 28, 2018/Accepted October 9, 2018/Published Online January 14, 2019

Abstract

The present study investigated the ground effects on the VIV of a closed box girder based on wind tunnel tests and numerical methods. First, we presented a vibration measurement system for wind-tunnel test to simulate ground effects on the bridge girder. Second, we investigated the Strouhal number (St), the VIV lock-in phenomena over a range of wind velocities, and the maximum VIV amplitudes and their change laws in correlation with the height from the ground in two different ground conditions. The outcomes showed the following: that a low height from the ground can lead to high St numbers; that ground effects did not change the width of VIV lock-in range, but did impel VIVs to occur earlier in the uniform flow field; and that the maximum VIV amplitudes was reduced and the VIV lock-in range narrowed in the turbulent flow field. Finally, we conducted the numerical simulation method to study the ground effect mechanism on VIVs. The numerical results showed that the periodic vortex shedding will generate periodic forces on the structure, which leads to the occurrence of VIVs. Ground effects can speed up vortex shifting, whereas aerodynamic forces on the deck are weakened.

Keywords: ground effects, numerical simulation, St number, Vortex-Induced Vibration (VIV), wind tunnel test

1. Introduction

Flow around bluff bodies typically results in the separation of the boundary layer and the roll-up of the separation shear layer, thereby forming an alternating vortex-shedding flow phenomenon. The fluctuating pressures around the bluff bodies generate nonlinear fluid–structure interactions, which also are known as Vortex-Induced Vibrations (VIVs). Although the VIV is a limited amplitude response, it is prone to occur at low wind speeds with relatively large amplitude, such as at the Xihoumen Suspension Bridge (Ge *et al.*, 2011), Rio-Niterói Bridge (Battista and Pfeil, 2000), Storebælt Suspension Bridge (Larsen *et al.*, 2000), and Trans-Tokyo Bay Bridge (Fujino and Yoshitaka, 2002). These vibrations affect traffic safety, influence strength and fatigue resistance of the structure, and may induce other types of instability such as cable vibrations. Therefore, in modern wind-resistant design for bridges, VIV performance design is weighed heavily and must be predicted accurately in the wind-resistance design of long-span bridges.

So far, scholars have conducted many investigations into bridge VIV in the unbounded flow field. A basic understanding of VIVs of bridge decks has improved over the past several decades, as it has been a subject of focused wind-tunnel experiments (Komatsu and Kobayashi, 1980; Diana *et al.*, 2006;

Schewe and Larsen, 1998; Ronaldo and Michèle, 2000; Larsen *et al.*, 2000; Zhiyong Zhou *et al.*, 2015; Fuyou Xu *et al.*, 2016; Yongxin Yang *et al.*, 2016; Wenyong Yuan *et al.*, 2017; Zhou *et al.*, 2017), full-scale observations (Kumarasena *et al.*, 1991; Owen *et al.*, 1996; Frandsen, 2001; Li *et al.*, 2011; Li *et al.*, 2014), and development of semi-empirical models (Larsen, 1995; Diana *et al.*, 2010; Teng Wu and Ahsan Kareem, 2013), as well as Computational Fluid Dynamics (CFD) schemes (Sarwar and Ishihara, 2010; Larsen and Walther 1997; Diana *et al.*, 2010; Sarwar and Ishihara, 2010; Larsen and Larsen, 2012). An excellent overview of VIVs of bridge decks is given in Teng Wu's studies (Teng Wu and Ahsan Kareem, 2012).

Currently, many existing and under-construction bridges, such as the Shenzhen-Hong Kong Channel Bridge, are built at low heights from the ground. Additionally, for those bridges located above reservoirs, as water level rises, the deck comes close to the water surface. In these cases, the air flow around the bridge girders, as a result of interactions with the ground or water surface, is definitely different from the conventional cases of bridges above a certain height. The formed channel between the deck and the ground inevitably will hinder or accelerate the airflow. This, in turn, will have an impact on the aerodynamic forces on the deck, which may result in unexpected wind-induced responses of the bridges. This phenomenon is regarded

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as “ground effects.”

To the best knowledge of the authors, no systematic studies into ground effects on the VIVs of the bridge have been performed. Studies of ground effects mainly have concentrated on aerospace engineering, in which case most of the research focuses on applied experimental approaches (Justin, 2012; Mei Yang, 2015; Marshall, 2010; Yang, 2014; Lu, 2014) and numerical simulations methods (Mokry, 2001; Barber, 2006; Park, 2008; Lee, 2010; Mahon, 2005; Su, 2013; Tang, 2013) to study the ground influences on the wings. Justin (2012) adopted experimental methods to study ground effects on NACA4412 airfoil under different wind attack angles and low Reynolds numbers. Mei Yang (2015) used the mobile road simulation system and the 6-component balance measuring system in wind tunnel tests to study the ground viscous effects on aerodynamic characteristics of the Wing-In-Ground (WIG) effect. Marshall (2010) and Yang (2014) studied the influence of boundary layer effects on the ground effects by the wind tunnel test method, and results showed that the thickness of the boundary layer reduced the effective height in the WIG wind tunnel tests so that ground effects became stronger.

In the study of the ground effects, the wind tunnel experiments are predominant, and the numerical simulations are relatively rare. Mokry (2001) adopted numerical simulation methods to study the ground effects on dissipation and movement of the trailing vortex of the aircraft. Studies showed that the ground effects accelerated the vortex dissipation. Barber (2006) used experimental and numerical simulation methods to study the ground effects on aerodynamic responses of the structure. Park (2008) adopted numerical simulation methods to study the ground effects on static and dynamic stability of the wing endplates, and explained the reasons for the variation of lift coefficients. The results showed that the ground effects decreased the aerodynamic stability of the wing endplates. Lee (2010) also applied numerical simulation methods to study the influence of wing configurations on aerodynamic characteristics.

In the CFD simulation, the Reynolds-Averaged Navier–Stokes (RANS) method and the Detached Eddy Simulation (DES) method are all widely used. The RANS method can simulate the mean flow field well at low cost, but its accuracy in the prediction of the primary vortex breakdown and the secondary vortex strength is not high (Yunpeng Qin, 2015). The DES method is an alternative to accurately capture the unsteady characteristics of various scale vortices at slightly higher cost. The DES and Delayed Detached Eddy Simulation (DDES) grounded on SST $k-\omega$ and Spalart–Allmaras (S–A) turbulence models are the most common numerical methods (Cummings, 2013; Yunpeng Qin, 2015 and Riou, 2015), among which the DDES works better for the massive or shallow separations (Cummings, 2013). But for the closed box girder, SST $k-\omega$ is efficient and accurate enough in both static and dynamic simulations.

When bridges are constructed at low heights from the ground, it is necessary to investigate the ground effects on the VIV performance and its influence mechanism. In this study, we have

adopted wind tunnel tests and numerical simulation methods to study the ground effects on the VIV of a closed box girder. The organization of this paper is as follows: Section 2 introduces an experimental setup to simulate ground effects on the closed box girder; Section 3 presents the Strouhal number (St), the VIV lock-in phenomena over a range of wind velocities, and the maximum VIV amplitudes and their change laws in correlation with the heights from the ground in two flow fields (two different ground roughness) through wind tunnel tests; Sections 4 and 5 analyze the mechanism of ground effects on vortex-excited resonance using numerical simulations, and Section 6 presents conclusions.

2. Vibration Measurement System for Wind Tunnel Tests of the Closed Box Girder

2.1 Section Model Design

A section model with high rigidity is required in the vibration measurement wind tunnel test. In this study, the model had a length L of 1.2 m, a height D of 0.047 m, and a width B of 0.3647 m, with a length-width ratio of 3.3. The designed model is shown in Fig. 1. The ground clearance H referred to the distance from the bottom surface of the model to the ground (uniform flow field) or the rough strip top surface (turbulent flow field). The rough strip layout and the position of the section model in the turbulent flow field are shown in Fig. 2. The side length d of each rough strip was 0.05 B , and the size of each rough strip was 0.018 m $\times$ 0.018 m $\times$ 2.4 m. We arranged up to 24 rough strips in this model, with a center distance of 0.108 m and a clear spacing of 0.9 m, equal to $5d$.

We conducted the wind tunnel tests in the TJ-2 wind tunnel in Tongji University. In the experiments, we created two different ground conditions, with and without rough strips, to simulate the turbulent flow field and the uniform flow field, respectively.

We conducted the vibration measurement tests under three different wind attack angles ($+3^\circ$, 0° , -3°) and four different heights from the ground in both flow fields (listed in Table 1). Case of $H/B = 4.0$ only exists in the uniform flow field and is considered as a case with no ground effects.

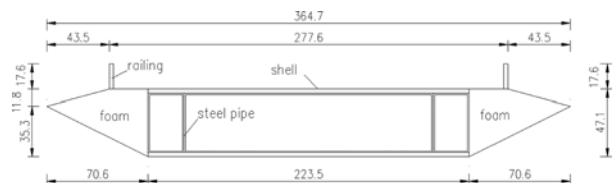


Fig. 1. Section Model for VIV (unit: mm)

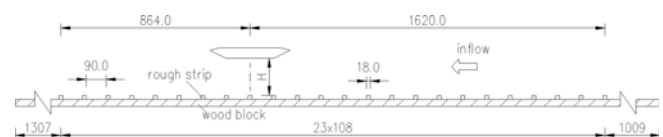


Fig. 2. Rough Strip Layout and the Position of Section Model: (a) without Ground Effects, (b) with Ground Effects

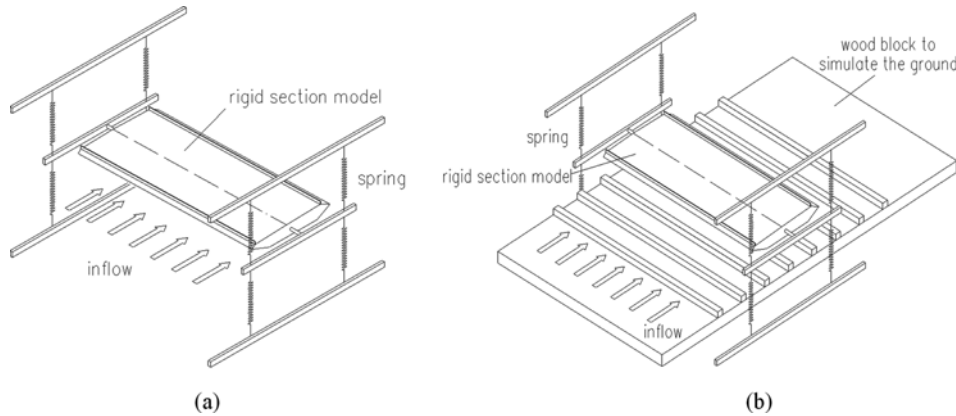


Fig. 3. Sketch of Experimental Installation: (a) Uniform Flow Field without Ground Effects, (b) Uniform Flow Field with Ground Effects, (c) Turbulent Flow Field with Ground Effects

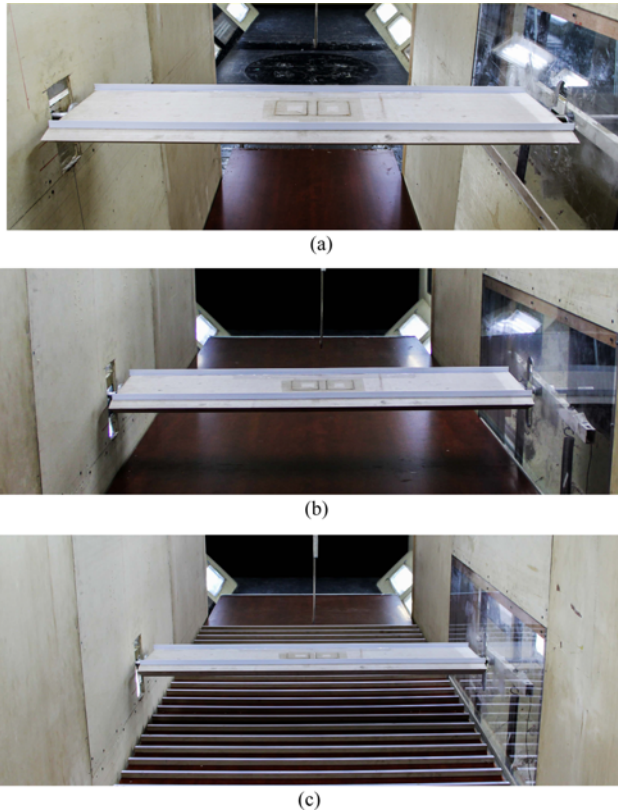


Fig. 4. Photos of Section Model VIV Measurement Tests

To measure the VIV lock-in ranges and the maximum amplitudes, we adopted the free oscillation method.

For the vibration measurement wind tunnel tests, we elastically suspended the rigid section model on a bracket in the wind tunnel using eight springs. Fig. 3 shows the section model installation diagram and Fig. 4 shows the test photos.

We achieved the desired vertical frequency and torsional frequency of the whole system by determining the appropriate mass and mass inertia of the entire system and adjusting the spring stiffness and spacing. We measured inflow speed by a pitot tube and micro-manometer located in front of the model.

Table 1. Cases of Section Model Wind-Induced Vibration Measurement Tests

Width B (mm)	Ground clearance H (mm)	H/B	Uniform flow field	Turbulent flow field
			Attack angle	Attack angle
364.7	72.9	0.2	$-3^\circ, 0^\circ, +3^\circ$	$-3^\circ, 0^\circ, +3^\circ$
	145.9	0.4		
	218.8	0.6		
	1458.8	4.0		

Table 2. Major Parameters of VIV Section Model Tests

	Parameters of VIV section model
Length of model L (m)	1.20
Height of model D (m)	0.047
Width of model B (m)	0.3647
Mass per length m (kg/m)	5.036
Vertical frequency f_h (Hz)	8.10
Mass moment per length I_m (kg·m ² /m)	-
Torsional frequency f_a (Hz)	-
Vertical damping ratio ζ_h	4.8‰
Torsional damping ratio ζ_a	-

Note: B = section width; m = FSI (Fluid Structure Interaction); D = section height, m; S_r = Strouhal number; L = length of the model, m; U = inflow velocity, m/s; H = ground clearance, m; VIV = Vortex-Induced Vibration; t = time, s; GE = Ground Effect; CFD = Computational Fluid Dynamics; I_m = mass moment per length, kg·m²/m; f = vortex-shedding frequency, α = wind attack angle

Generally, VIV is sensitive to the damping ratio of the system. For this reason, before performing wind tunnel tests, we measured the damping ratio of the system, which is given in Table 2. Because VIV is likely to occur at low wind speeds, we selected a set of springs with higher rigidity to improve the vertical frequency of the overall system. This study concentrated on vertical VIV, so it was necessary to ensure that the torsional frequency was far away from the vertical frequency to avoid the mutual influence between vertical VIV and torsional VIV. Major parameters of the VIV section model wind tunnel tests are

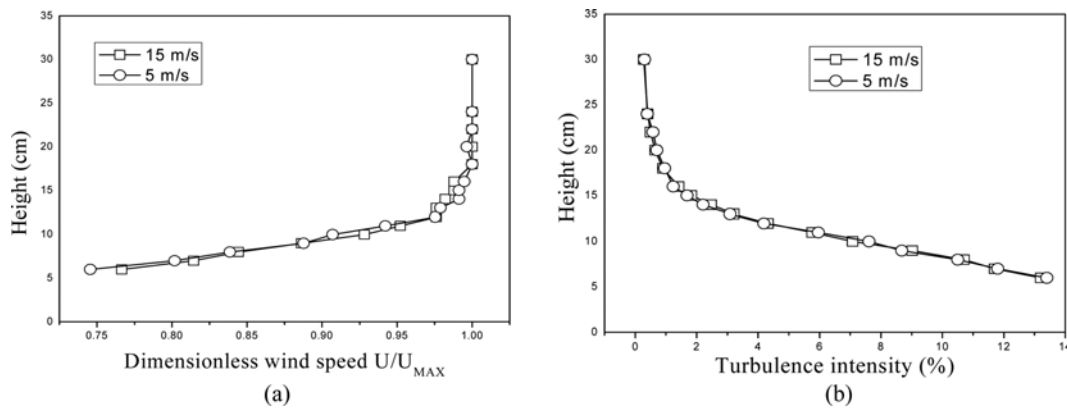


Fig. 5. The Average Wind Profiles and Turbulence Profile Test Results: (a) Mean Wind Profile, (b) Turbulence Intensity Profile

shown in Table 2.

2.2 Wind Profile and Turbulence Intensity Profile of Turbulence Flow Field

Before the vibration measurement tests, it was essential to achieve a stable turbulent flow field. Therefore, in the absence of a section model, 16 measuring points at different heights from the ground were arranged vertically in a row along the centric position of the section model (The distances from the measuring point to the top of rough strips were 6 cm, 7 cm, 8 cm, 9 cm, 10 cm, 11 cm, 12 cm, 13 cm, 14 cm, 15 cm, 16 cm, 18 cm, 20 cm, 22 cm, 24 cm, and 30 cm.). We monitored and measured the reference wind speeds with a pitot tube and micro-pressure gauge. We collected wind speeds and turbulence data using a Cobra Dimensional Fluctuating Wind Speed Measuring Instrument (Turbulent Flow Instrumentation, Victoria, Australia).

Figure 5 presents the results of mean wind profiles and turbulence intensity (fluctuating velocity/ average velocity) profiles under 2 inflow speeds of 5 m/s and 15 m/s in the turbulent flow field, which are considered capable of providing stable wind field profiles for the subsequent wind tunnel tests.

It could be inferred from the results that the arrangement of rough strips formed a certain thickness of the wind profile and turbulence intensity profile. The height of the profile was about 15–20 cm, which was equivalent to the height from the ground $H = (0.4 \sim 0.5) B$.

3. Test Results of Ground Effects on VIV

3.1 Ground Effects on the St Number

For VIV, St number is a dimensionless coefficient reflecting shedding frequencies of the vortex. From the formula $St = fD/U$, it is obvious that when the wind speed is at a certain value, a larger St number indicates a higher frequency of vortex shedding.

Take the case of 0° wind attack angle as a representation. The St number variation curves with the height from the ground is shown in Fig. 6, where f is the vortex-shedding frequency, D is the section height (0.047 m), and U is the inflow velocity (5 m/s). In the following analyses, the ground clearance of $H/B = 4$ represents

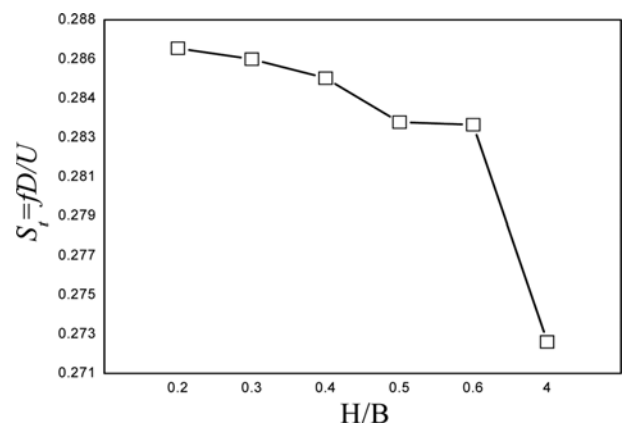


Fig. 6. St Number Test Values in Uniform Flow Field with Wind Attack Angle of 0°

the unbounded flow field. Fig. 6 reveals that under the constant wind speed in the uniform flow field, when the ground clearance decreases, the St number grows nonlinearly, and, as mentioned previously, the vortex-shedding frequency becomes larger.

For a specific section, the oscillating frequency of a certain VIV normally is constant. Seen from the formula $St = fD/U$ again, when f and D are constant, a higher St means a lower wind speed U . In other words, a lower height from the ground can lead to higher St numbers, which results in the vortex-shedding frequency meeting the oscillating frequency of the structure at a lower wind speed and, finally, VIV occurs. Thus, ground effects will impel VIVs to occur earlier.

3.2 Ground Effects on VIV Lock-in Range and Amplitudes

Section model test results of vertical VIV displacements are shown in Fig. 7:

- 1) In the uniform flow field, ground effects do not change the width of VIV lock-in range, but they do impel VIVs to occur earlier, with wind speed intervals shifting forward about 1 m/s. This indicates that ground effects will speed up vortex shedding, thus making the VIV occur in advance. This conclusion is in accordance with results in Section 3.1.
- 2) In the turbulent flow field, because VIV is more sensitive to

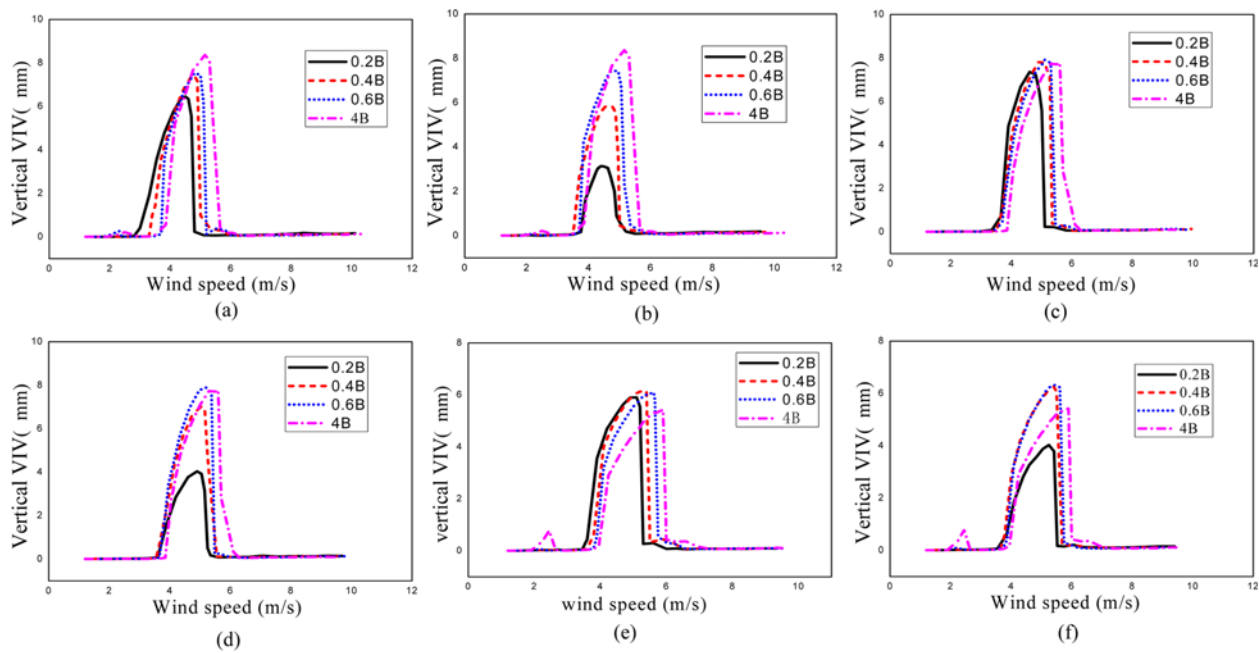


Fig. 7. Responses of Vertical VIV under Two Kinds of Flow Field, Different Wind Attack Angles and Heights from the Ground: (a) Uniform Flow Field, $+3^\circ$ Wind Attack Angle, (b) Turbulent Flow Field, $+3^\circ$ Wind Attack Angle, (c) Uniform Flow Field, 0° Wind Attack Angle, (d) Turbulent Flow Field, 0° Wind Attack Angle, (e) Uniform Flow Field, -3° Wind Attack Angle, (f) Turbulent Flow Field, -3° Wind Attack Angle

turbulence, a lower height from the ground means higher turbulence, so that vibration amplitude is reduced and the VIV lock-in range is narrowed (see Figs. 7(b), 7(d) and 7(f)).

- 3) Ground effects on VIV amplitudes vary at different wind attack angles. At the wind attack angle of $+3^\circ$, ground effects reduce VIV amplitudes. In the case of -3° , ground effects have little impact on VIV amplitudes. At the wind attack angle of 0° , ground effects slightly increase the amplitude, but the influence is not significant.
- 4) The VIV lock-in range and amplitudes are almost the same when the height H equals $0.6B$. This phenomenon appears in the same wind attack angle in both flow fields. This means that when the height from the ground is over a certain value, ground roughness has little effect on VIV.

4. Numerical Simulation for Ground Effects on VIV

4.1 Computational Domain and Grid

In this study, we adopted Finite Volume Method (FVM) and

Computational Fluid Dynamics (CFD) to simulate two cases of ground effects on the VIV of the closed box girder: 1) CASE 1, girder at the height of $H/B = 0.2$ under $+3^\circ$ wind attack angle in the uniform flow, and 2) CASE 2, girder at the height of $H/B = 4$ under $+3^\circ$ wind attack angle in the uniform flow. The simulation mainly focuses on the smooth ground effects. Computations were performed by the software FLUENT.

Due to the complexity of turbulence, it may require 3D numerical approaches to analyze the synthetic influence of both the ground and turbulence on the wind-induced responses of bridge decks. We will discuss it in the future.

We completed mesh generation using the software ANSYS ICEM CFD. To avoid failure in computations resulting from mesh distortions and negative volumes, similar to the approach proposed by Fransos and Bruno (2006), we discretized the domain into a rigid boundary layer mesh region, dynamic mesh region, and stationary mesh region, as depicted in Fig. 8, where the ground clearance H is equal to $0.2B$ (in CASE 1) or $15h$ (in CASE 2), h is the deck height, and B is the deck width.

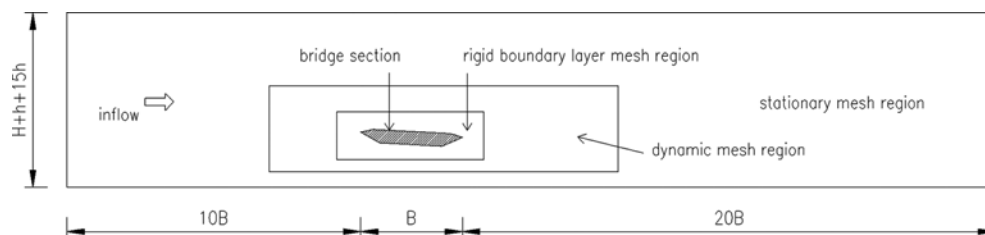


Fig. 8. Overview of Computational Domain: (a) $H/B = 0.2$, $+3^\circ$ Wind Attack Angle, (b) $H/B = 4$, $+3^\circ$ Wind Attack Angle, (c) Boundary Layer Grids

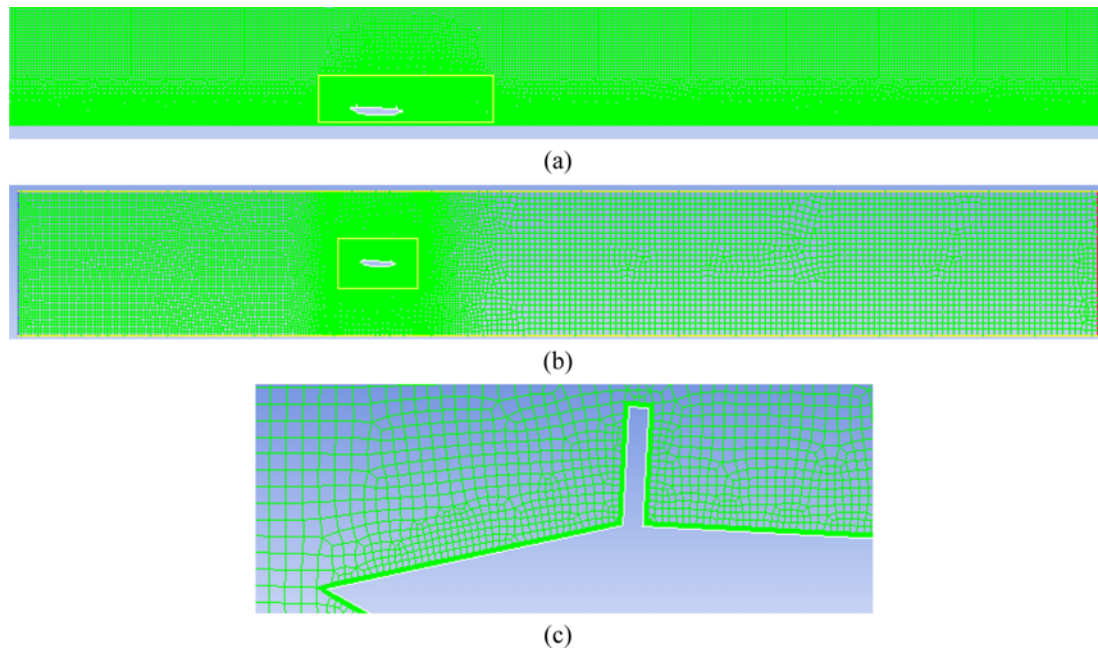


Fig. 9. Sketch of Computing Grids

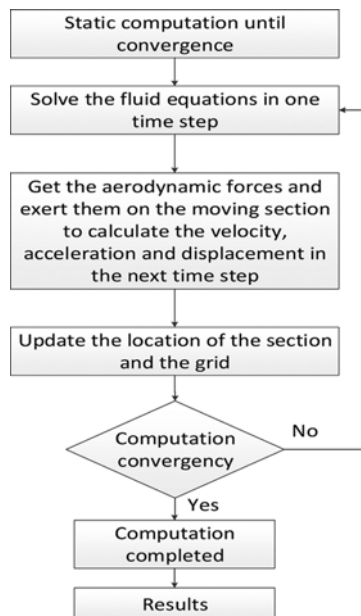


Fig. 10. Computation Procedure of FSI

We generated body-fitted unstructured quadrilateral grids in the rigid boundary layer mesh region. The thickness of the first layer cells is 0.0001 m to ensure simulation accuracy. This region had the same motion as the bridge section, without smoothing and remeshing in the process. We filled the dynamic mesh region with triangle grids so that it could be updated during the computation. Quadrilateral and triangular grids covered the stationary mesh region, which remained stationary in the entire computation process. Sketches of the computing grids for $H/B = 4$ and $H/B = 0.2$ are shown in Fig. 9.

4.2 Turbulence Model and Boundary Conditions

The upstream boundary was set as velocity-inlet, with different velocity magnitudes in the x direction and zero in the y direction. The turbulence intensity and the turbulence viscosity ratios (turbulent viscosity/dynamic viscosity) were 0.5% and 10%, respectively. The outflow boundary was set as pressure-outlet, with a gauge pressure of 0. The upper boundary was set as symmetry and the lower boundary was set as no slip wall in CASE 1 and symmetry in CASE 2. The section surface had no slip wall condition.

We selected SIMPLEC (Semi-Implicit Method for Pressure Linked Equations Consistent) algorithm as the pressure-velocity coupling scheme and the time step size was 0.0002 s. We used the SST $k-\omega$ turbulence model and adopted the transient pressure-based solver.

4.3 Fluid–Structure Interaction (FSI) Modeling

In modeling, the equations of fluid and motion of the structure must be solved simultaneously. Hence, to solve both equations, a moving grid is needed at each time step for the fluid portion. First, the structure is fixed for static computation of flow around the deck. Second, when the computation convergence is gained, the structure then is released for FSI computation. In this study, specific work on the FSI computation was as follows:

- 1) Get the aerodynamic forces of the structure by the macro `Compute_Force_And_Moment` provided in the user-defined function of FLUENT.
- 2) Adopt the *Newmark- β* method to calculate the structure responses, namely, the velocity, acceleration and displacement.
- 3) Define the velocity, acceleration and displacement of the structure by the macro `DEFINE_CG_MOTION`, and then

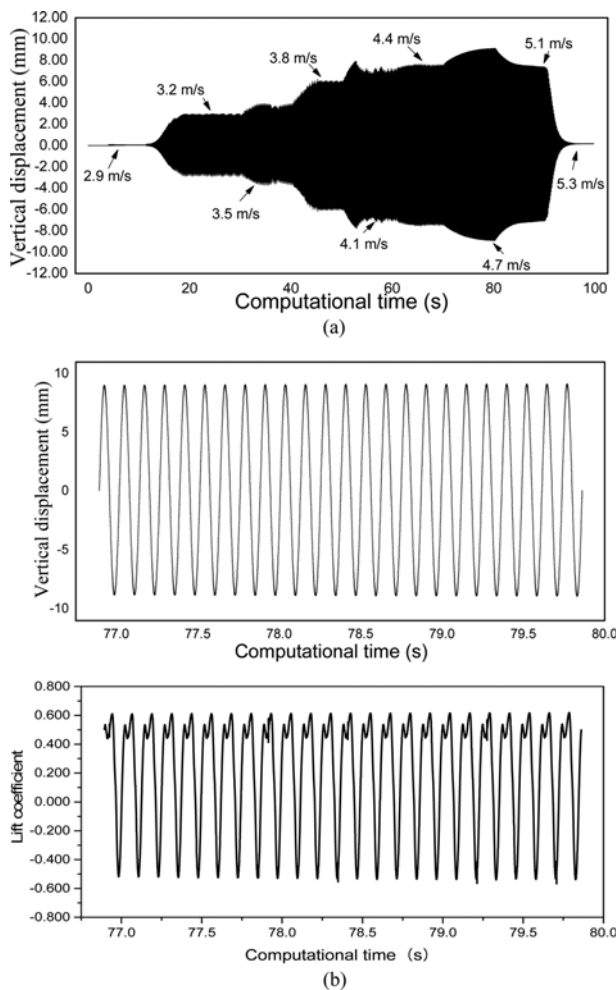


Fig. 11. Vertical Displacement Curves and Lift Coefficient Curves: $H/B = 4$: (a) Vertical Displacement Curves with Computational Time: $H/B = 0.2$, $+3^\circ$ Wind Attack Angle, Uniform Flow, (b) Vertical Displacement and Lift Coefficient Curves with Computational Time, Inflow Speed 4.7 m/s: $H/B = 0.2$, $+3^\circ$ Wind Attack Angle, Uniform Flow

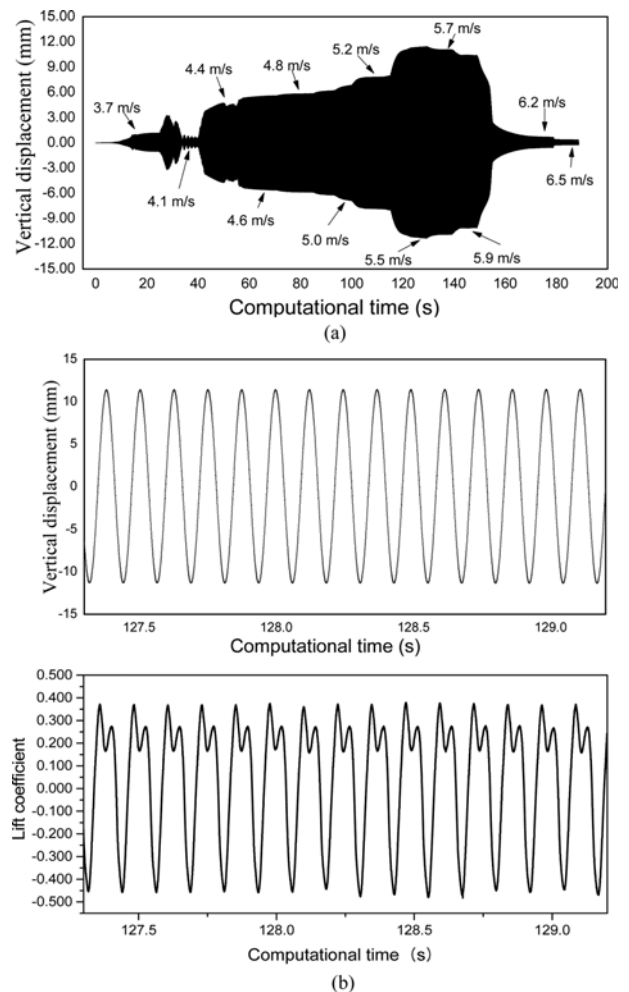


Fig. 12. Vertical Displacement Curves and Lift Coefficient Curves: $H/B = 4$: (a) Vertical Displacement Curves with Computational Time: $H/B = 4$, $+3^\circ$ Wind Attack Angle, Uniform Flow, (b) Vertical Displacement and Lift Coefficient Curves with Computational Time, Inflow Speed 5.5 m/s: $H/B = 4$, $+3^\circ$ Wind Attack Angle, Uniform Flow

update the location of the structure and the grid.
The computation procedure of FSI is given in Fig. 10.

4.4 Computational Results of Ground Effects on VIV Lock-in Range and Amplitudes

Major parameters in the numerical simulations were in accordance with corresponding cases in the wind tunnel tests. For CASE 1 ($H/B = 0.2$), the range of computational velocity was from 2.9 m/s to 5.3 m/s, for a total of 9 inflow speeds. For CASE 2 ($H/B = 4$), the range of computational velocity was from 3.7 m/s to 6.5 m/s, for a total of 12 inflow speeds. The vertical displacement curves and the lift coefficient curves with computational time are presented in Figs. 11 and 12.

Figure 13 gives responses of vertical VIV for the calculated values and the wind tunnel test values. Seen from the results, the computational results agree well with the tests results. Thus, it is convincing to analyze the mechanism of ground effects on VIV

by displaying the visible flow field from the computation.

5. Mechanism of Ground Effects on VIV

Because VIV is a limited motion amplitude phenomenon, when the displacement comes to a steady state, so does the interaction of the flow field and the structure. Therefore, it is meaningful to analyze the flow fields in one period (T). For CASE 1 and CASE 2, to study the mechanism, we selected 4 positions of each case, in which the VIV amplitude reaches the maximum with inflow speed of 4.7 m/s for CASE 1 and 5.5 m/s for CASE 2, $t_1 = 0$ T, $t_2 = 1/4$ T, $t_3 = 2/4$ T, and $t_4 = 3/4$ T (as shown in Fig. 14). Contours of vorticity and pressure are shown in Figs. 14 and 15.

Figure 14 indicates that great resemblance existed in the instantaneous flow fields of the two cases. The flow around the moving section separated at the windward-side railing and formed

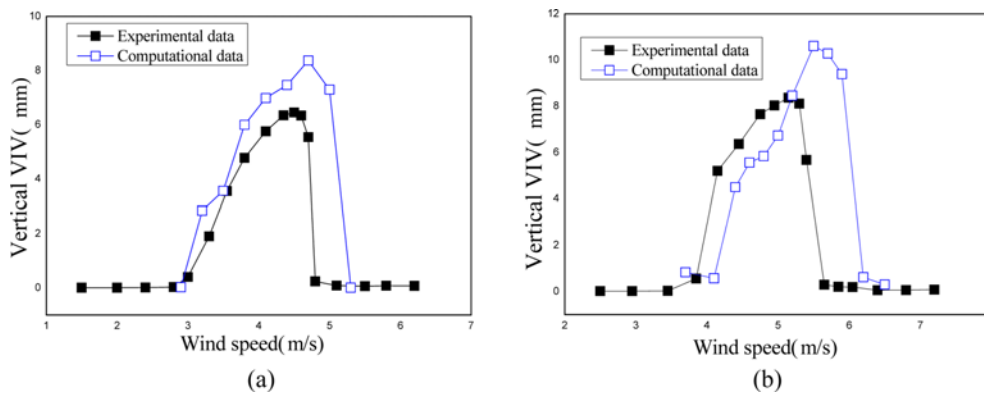


Fig. 13. Responses of Vertical VIV of the Calculated Values and the Wind Tunnel Test Values: (a) $H/B = 0.2$, $+3^\circ$ Wind Attack Angle, Uniform Flow, (b) $H/B = 4$, $+3^\circ$ Wind Attack Angle, Uniform Flow

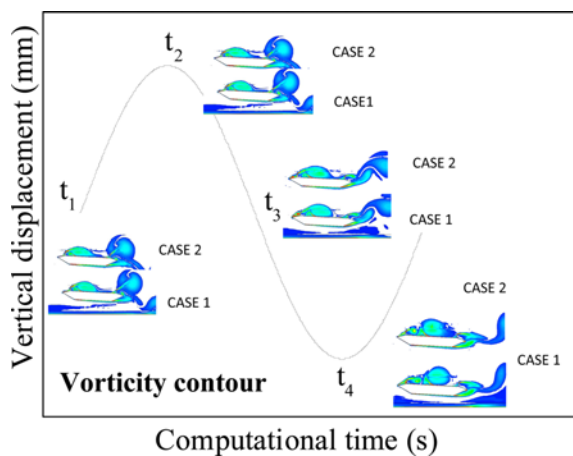


Fig. 14. Different Time Points and Vorticity Contours of Time $t_1 = 0$, $t_2 = 1/4T$, $t_3 = 2/4T$ and Time $t_4 = 3/4T$ in the Uniform Flow Field at $+3^\circ$ Wind Attack Angle for CASE 1 and CASE 2

a strong periodic vortex-shedding and -shifting phenomenon. As is known to all, one vortex means one lifting-surface, and vortex shifting means lifting-surface shifting. This periodic vortex shedding and shifting generated periodic forces on the structure, which led to the occurrence of VIVs (see Fig. 15).

Figure 15 reveals that ground effect impelled the velocity of the vortex shifting, and increased the St number because the existence of a boundary layer generally reduces the flow speed (in comparison with inflow speed) in the region between the deck and the ground, while enhancing the flow separation in the upper surface separation point. This accelerated the upper vortex shedding and shifting, thus making VIV occur earlier. This conclusion is in accordance with the results of ground effects on the St number in Section 3.1.

As is known to all, the oscillating force that acted on the box girder by this alternate vortex shedding increased linearly following an increase in the strength of vortices, though it decreased quadratically with an increase in the distance between the vortex core and the surface of the box girder. Ground effects increased the distance of the lifting surface induced by the vortex from the bridge section and weakened the strength of the vortex and thus

decreased the aerodynamic forces, which was reflected in the lift coefficient in Figs. 11(b) and 12(b), and the VIV amplitude was reduced. This conclusion agrees with those in Section 3.2 (see Fig. 7).

6. Conclusions

This study aimed to adopt the wind tunnel tests and numerical simulation methods for further studying the ground effects on the VIV of a closed box girder. A vibration measurement system was presented for wind-tunnel test in order to simulate the ground effects on the bridge girder is presented. Next, the Strouhal number, the vortex-induced vibration lock-in phenomena over a range of wind velocities, and the maximum VIV amplitudes and their change laws in correlation with the heights from the ground in two flow fields were studied. Important conclusions are as follows:

1. When the height from the ground decreased, the St number grows, and the vortex-shedding frequency became larger. In other words, a lower height from the ground can lead to higher St numbers, which produces the result that the vortex-shedding frequency meets the oscillating frequency of the structure at a lower wind speed and finally VIV occurs.
2. In the uniform flow field, ground effects did not change the width of the VIV lock-in range, but did impel VIVs to occur earlier.
3. In the turbulent flow field, because VIV was more sensitive to turbulence, a lower height from the ground meant higher turbulence, so that vibration amplitude was reduced and the VIV lock-in range was narrowed.
4. At the wind attack angle of $+3^\circ$, ground effects reduced VIV amplitudes. In the case of -3° , ground effects had little impact on VIV amplitudes. At the wind attack angle of 0° , ground effects slightly increased the amplitude, but this impact was not significant.
5. The VIV lock-in range and amplitudes were almost the same when the height H equaled $0.6 B$. This means that when the height from the ground was over a certain value, ground roughness had little effect on VIV.

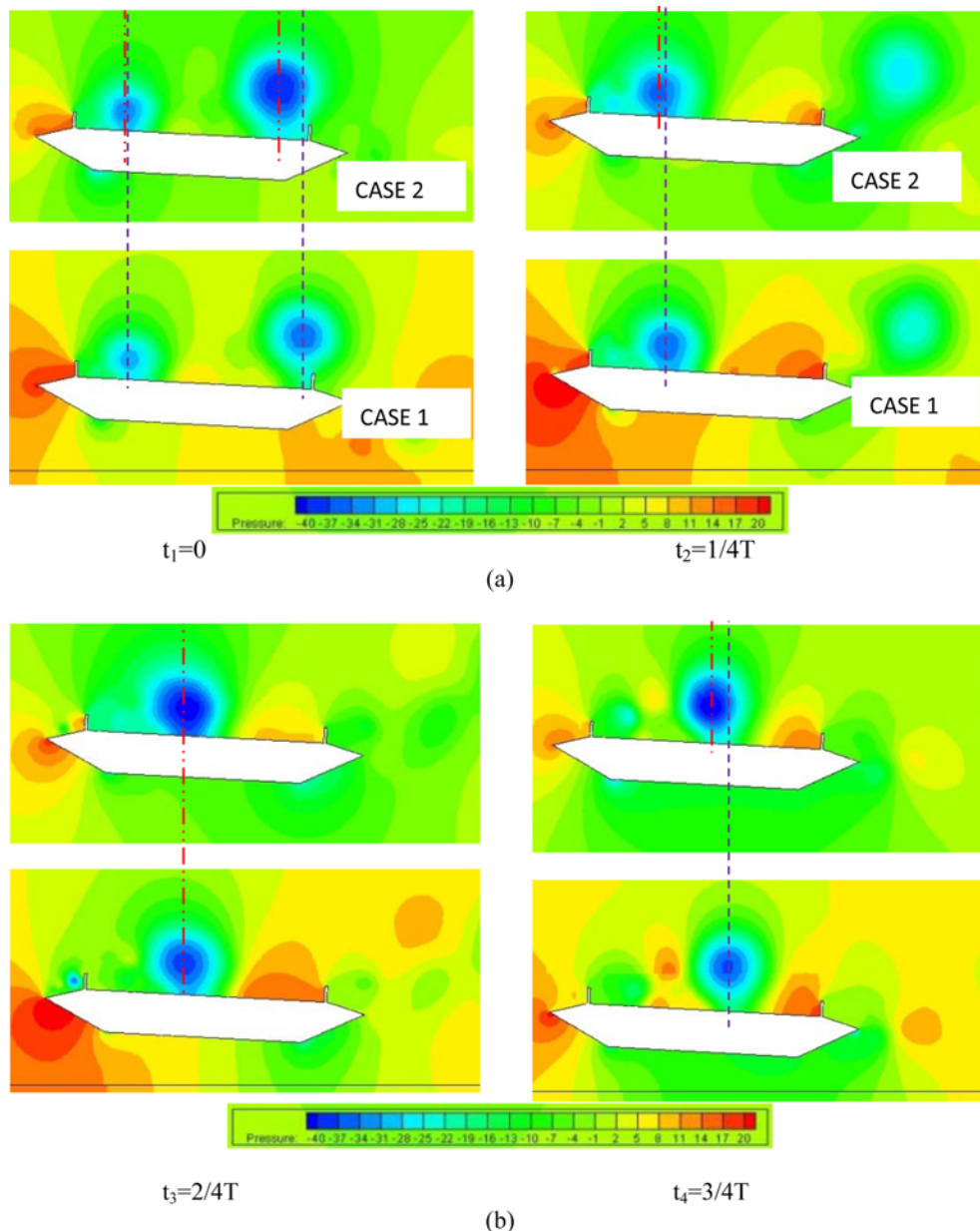


Fig. 15. (a) Pressure Contours of Time $t = 0$ and Time $t = 1/4T$ in the Uniform Flow Field at $+3^\circ$ Wind Attack Angle for CASE 1 and CASE 2, (b) Pressure Contours of Time $t_3 = 2/4T$ and Time $t_4 = 3/4T$ in the Uniform Flow Field at $+3^\circ$ Wind Attack Angle for CASE 1 and CASE 2

6. The flow around the moving section separated at the windward-side railing and formed a strong periodic vortex-shedding and -shifting phenomenon, which generated periodic forces on the structure and led to the occurrence of VIVs.
7. Ground effects impelled the vortex shedding and shifting, and increased the St number because the existence of boundary layer generally reduced the flow speed in the region between the deck and the ground, while increasing the flow speed (in comparison with inflow speed) in the upper surface. This accelerated the upper vortex shifting, thus making the VIV occur earlier.
8. Ground effects increased the distance of the lifting surface

induced by the vortex from the bridge section and weakened the strength of the vortex and thus decreased the aerodynamic forces, which reduced the VIV amplitude.

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